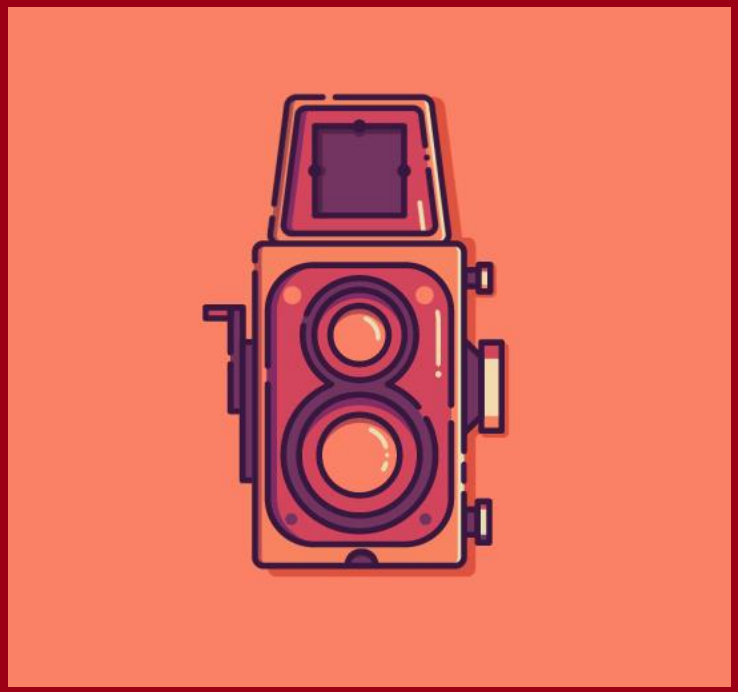
Università degli Studi di Padova

DIPARTIMENTO DI INGEGNERIA DELL'INFORMAZIONE



Digital Forensics & Biometrics

A.A. 2026/2027

Lecture 09 – camera forensics

Simone Milani
Room 216
DEI A

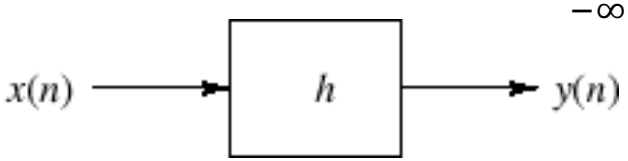
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simone.milani@dei.unipd.it

ANALYSIS OF SIGNALS: FILTERING

Filtering is widely used on biometric signals

$$y(t) = \int_{-\infty}^{\infty} x(a)\, h(t-a)da$$

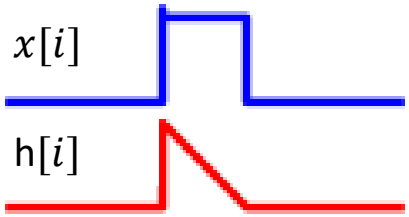
$$y[n] = \sum_{i=0}^{N} h[n-i]\, x[i] = h * x[n]$$

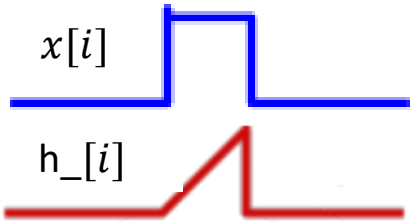


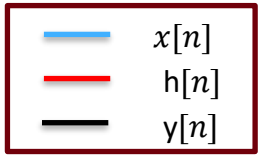
where h[n] is the response of the filter to the impulsive input

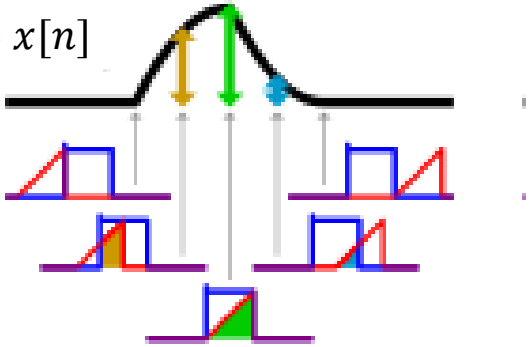
$$\delta[n] = \begin{cases} 1 & n = 0 \\ 0 & n \neq 0 \end{cases}$$

Ideally, it is like computing the correlation between x$[i]$ and $h_[i]$ with delay n

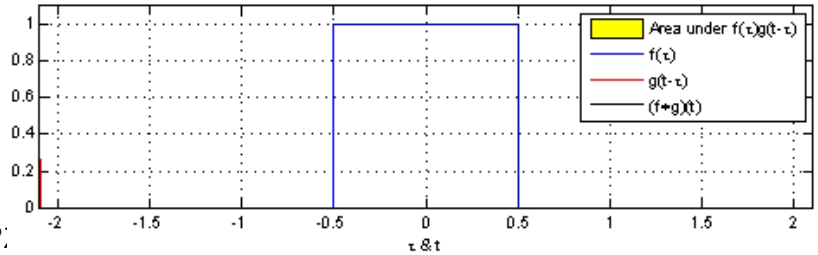








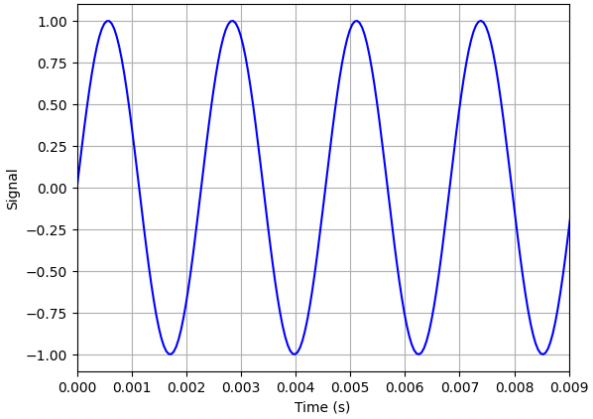
Example:
convolution of two rect



filters catch some specific structure in a signal

EXAMPLE: 1D SIGNALS (AUDIO)

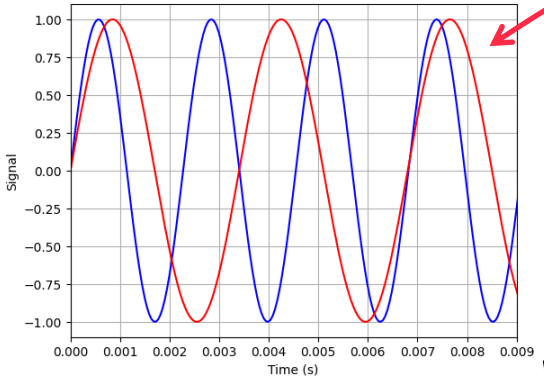
- Example from sounds but it can be applied to different things (typing patterns, gait, …)
- Consider a simple signal $x_0 = \sin(2\pi f_0 t)$ with $f_0 = 440\ Hz$. (C note)



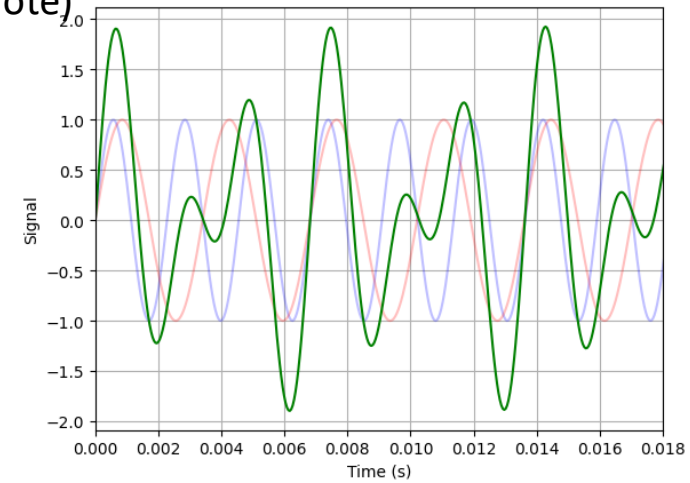


- Adding another note makes it more complex $x_0 + x_1$

$x_1 = \sin(2\pi f_1 t)$ with $f_1 = 294\ Hz$. (D note)



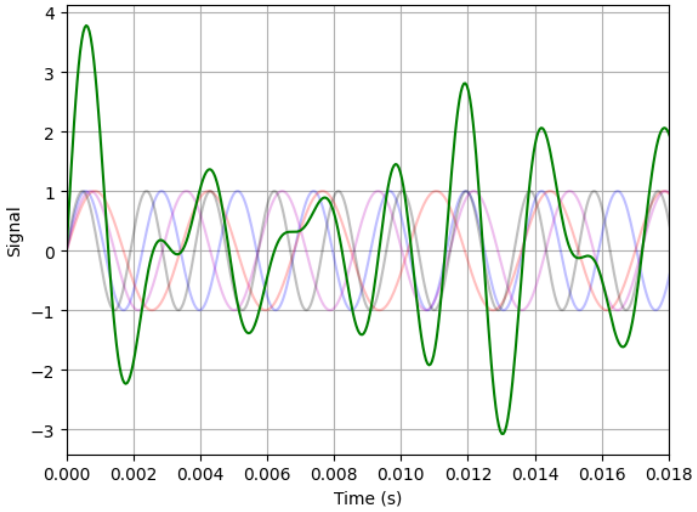


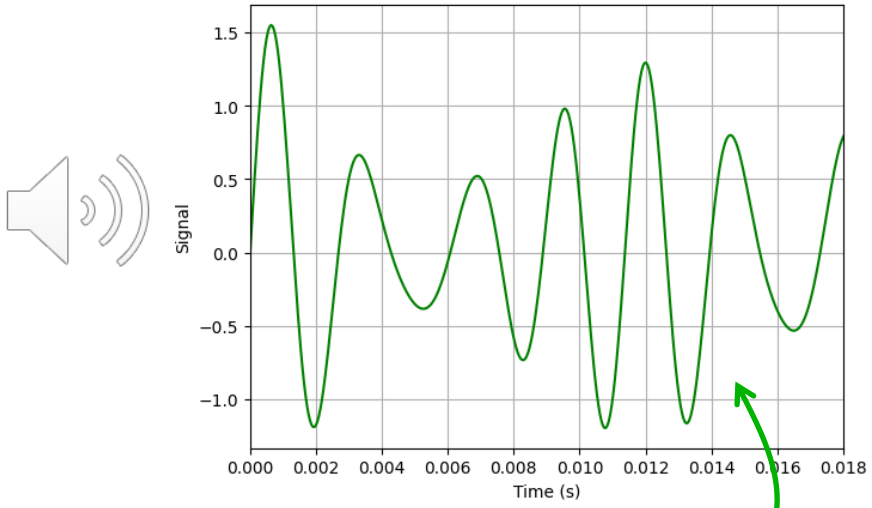




EXAMPLE: 1D SIGNALS (AUDIO)

- Itrìerate multiple frequencies $x_0 + x_1 + x_2 + x_3$
  with $x_i = \sin(2\pi f_i t)$ with $f_2 = 349\ Hz$ and $f_2 = 523\ Hz$





- Any signal can be decomposed as a combination of simple sinusoidal components (with different amplitudes

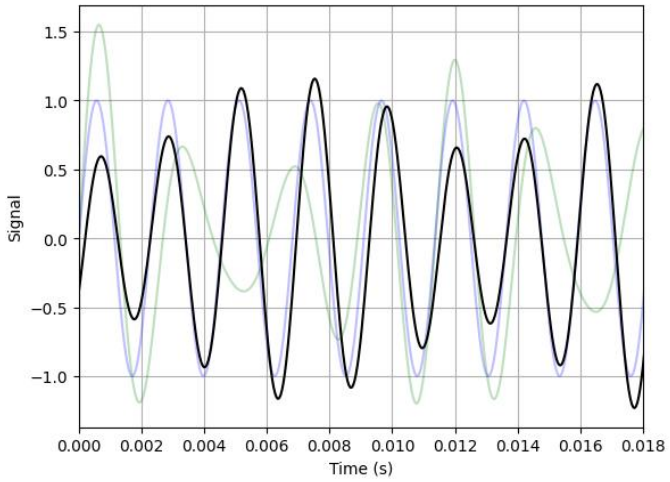
$$0.5\ x_0 + 0.2\ x_1 + 0.8\ x_2 + 0.4\ x_3$$

- How to compute the right combination givena a signal –> Fourier transform

FOURIER TRANSFORM

- Fourier transform can be seen as a filtering with sine-cosine signals
- Example: $0.5\,x_0 + 0.2\,x_1 + 0.8\,x_2 + 0.4\,x_3$

filter with $\mathrm{h(t)} = \sin(2\,\pi f_0 t)$ get the coefficient 0.5



- Filter with complex functions $\cos(2\,\pi f_0 t) + j\sin(2\,\pi f_0 t) = e^{-j2\,\pi f_0 t}$ and change the frequency f_0 in order to uncover all the weights (coefficients)
- Complete representation of the signal

FOURIER TRANSFORM

☐ Continuous-Time Fourier Transform (CTFT)

referred as Fourier spectrum or simply spectrum

$$\omega = 2\pi f$$

$$X_a(f) = \int_{-\infty}^{\infty} x_a(t) e^{-j\omega t} dt = \int_{-\infty}^{\infty} x_a(t) e^{-j2\pi f t} dt$$

inverse CTFT $$x_a(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X_a(f) e^{j2\pi t f} df$$

☐ Discrete Fourier Transform (DFT). (assuming *x* is defined by an array of N values)

$$X(k) = \sum_{n=0}^{N-1} x[n] e^{-j2\pi \frac{kn}{N}} = |X(k)| e^{j\theta(k)}$$

Magnitude (dB) $|X_a(f)|$

Phase (rad) $\theta(f) = \arg\{X_a(f)\}$

Inverse Discrete Fourier Transform (IDFT)

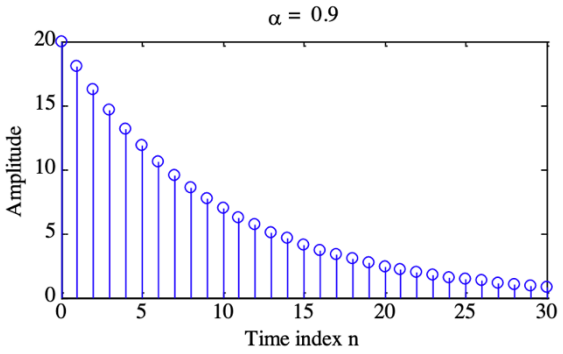
$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{j2\pi \frac{kn}{N}}$$

An original perspective https://www.youtube.com/watch?v=spUNpyF58BY

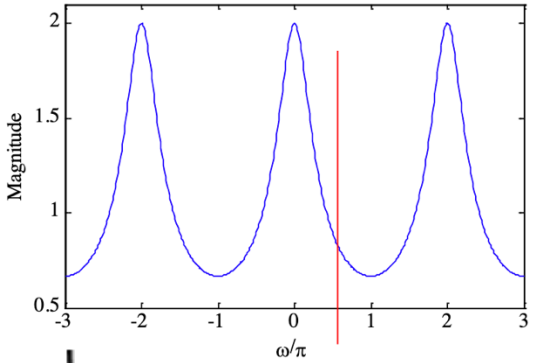
FOURIER PROPERTIES

Operation	$f(t)$	$F(\omega)$
Addition	$f_1(t) + f_2(t)$	$F_1(\omega) + F_2(\omega)$
Scalar multiplication	$kf(t)$	$kF(\omega)$
Symmetry	$F(t)$	$2\pi f(-\omega)$
Scaling (a real)	$f(at)$	$\frac{1}{\lvert a\rvert} F\left(\frac{\omega}{a}\right)$
Time shift	$f(t - t_0)$	$F(\omega)e^{-j\omega t_0}$
Frequency shift (ω_0 real)	$f(t)e^{j\omega_0 t}$	$F(\omega - \omega_0)$
Time convolution	$f_1(t) * f_2(t)$	$F_1(\omega)F_2(\omega)$
Frequency convolution	$f_1(t)f_2(t)$	$\frac{1}{2\pi} F_1(\omega) * F_2(\omega)$
Time differentiation	$\frac{d^n f}{dt^n}$	$(j\omega)^n F(\omega)$
Time integration	$\int_{-\infty}^{t} f(x)\,dx$	$\frac{F(\omega)}{j\omega} + \pi F(0)\delta(\omega)$

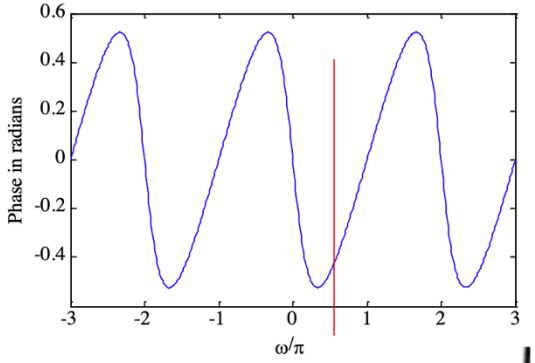
FOURIER TRANSFORM: 1D EXAMPLES



$|X(e^{j\omega})|$



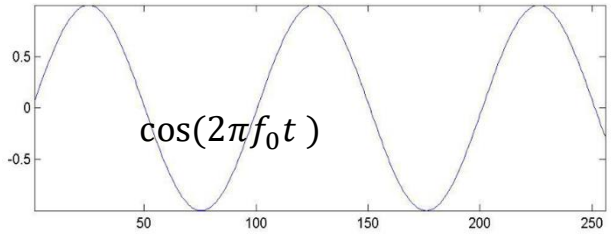
$\theta(\omega) = \arg\{X(e^{j\omega})\}$

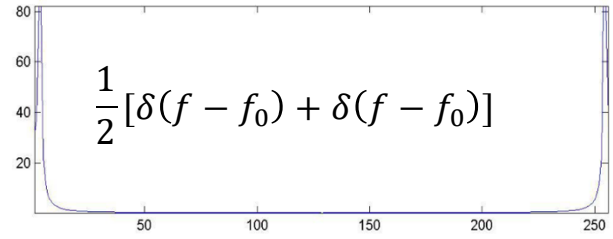


$x[n] = \alpha^n \mu[n], \quad |\alpha| < 1$

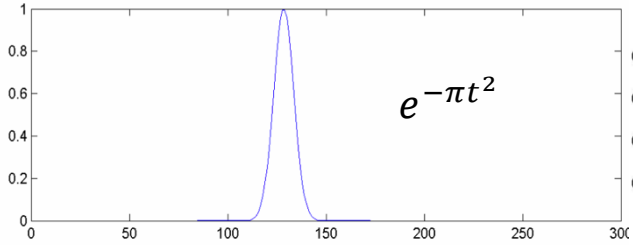
$$X(e^{j\omega}) = \frac{1}{1-0.5e^{-j\omega}}$$

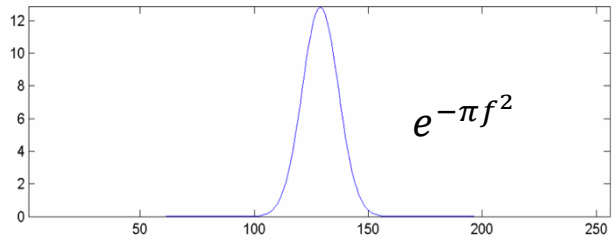
sin/cos



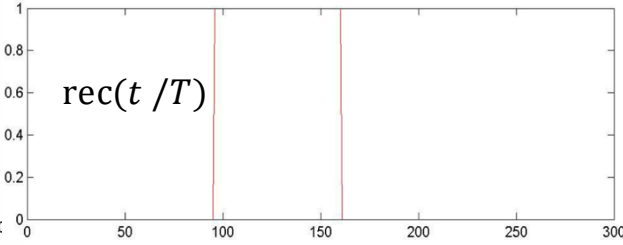


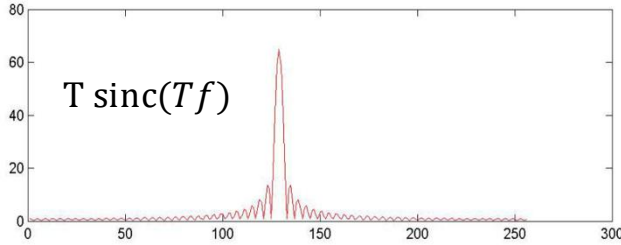
gaussian



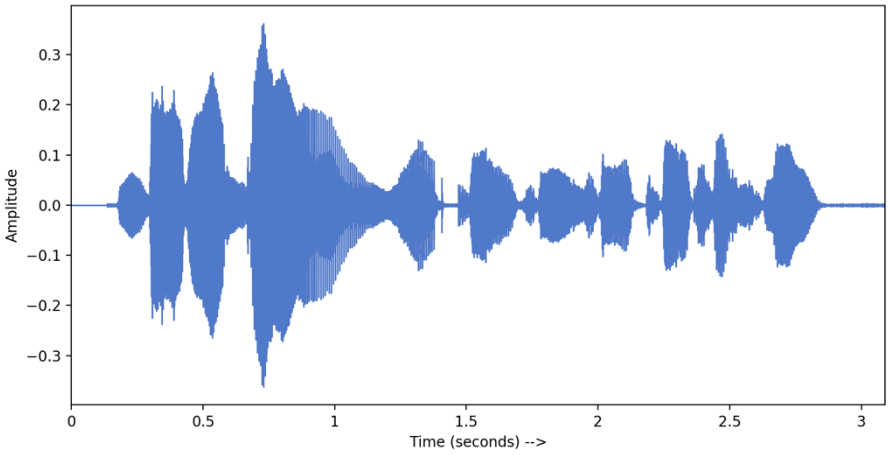


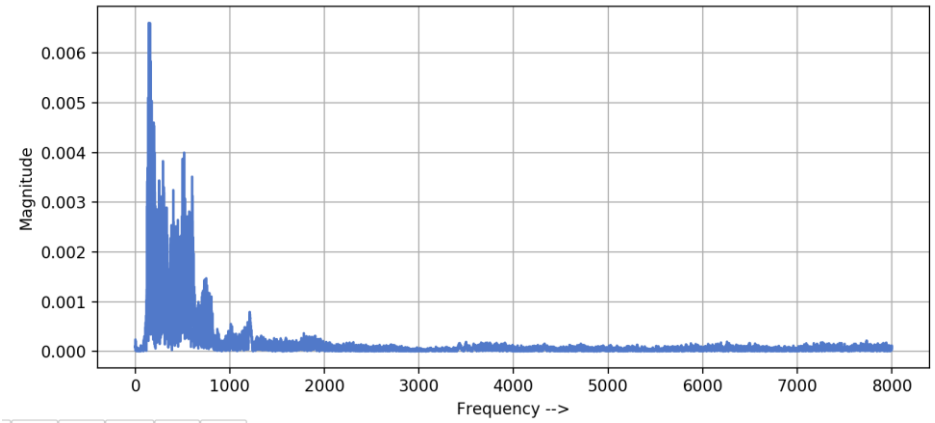
rect

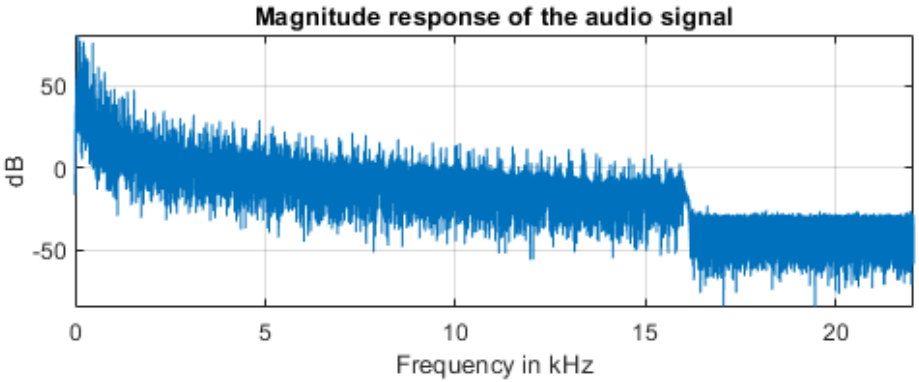




FOURIER TRANSFORM: 1D EXAMPLES



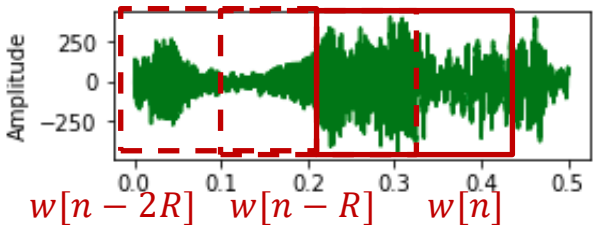




SPECTROGRAMS

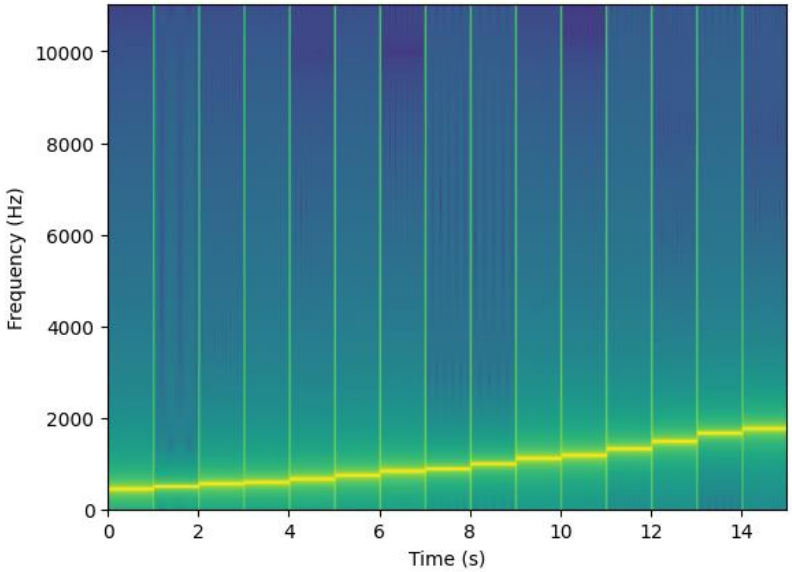
Signals can be represented by spectrograms using Short-time (or term) Fourier Transform (STFT)

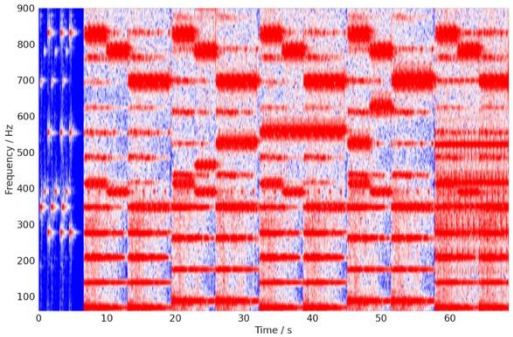
$$X_d(f) = \sum_{n=-\infty}^{\infty} x[n]w[n - dR]e^{-j2\pi fnT}$$



If Constant-Overlap-Add (COLA) propetry is verified $\sum_{d=-\infty}^{\infty} w[n - dR] = 1. \forall n\,.$ we have $\sum_{d=-\infty}^{\infty} X_d(f) = X(f)$

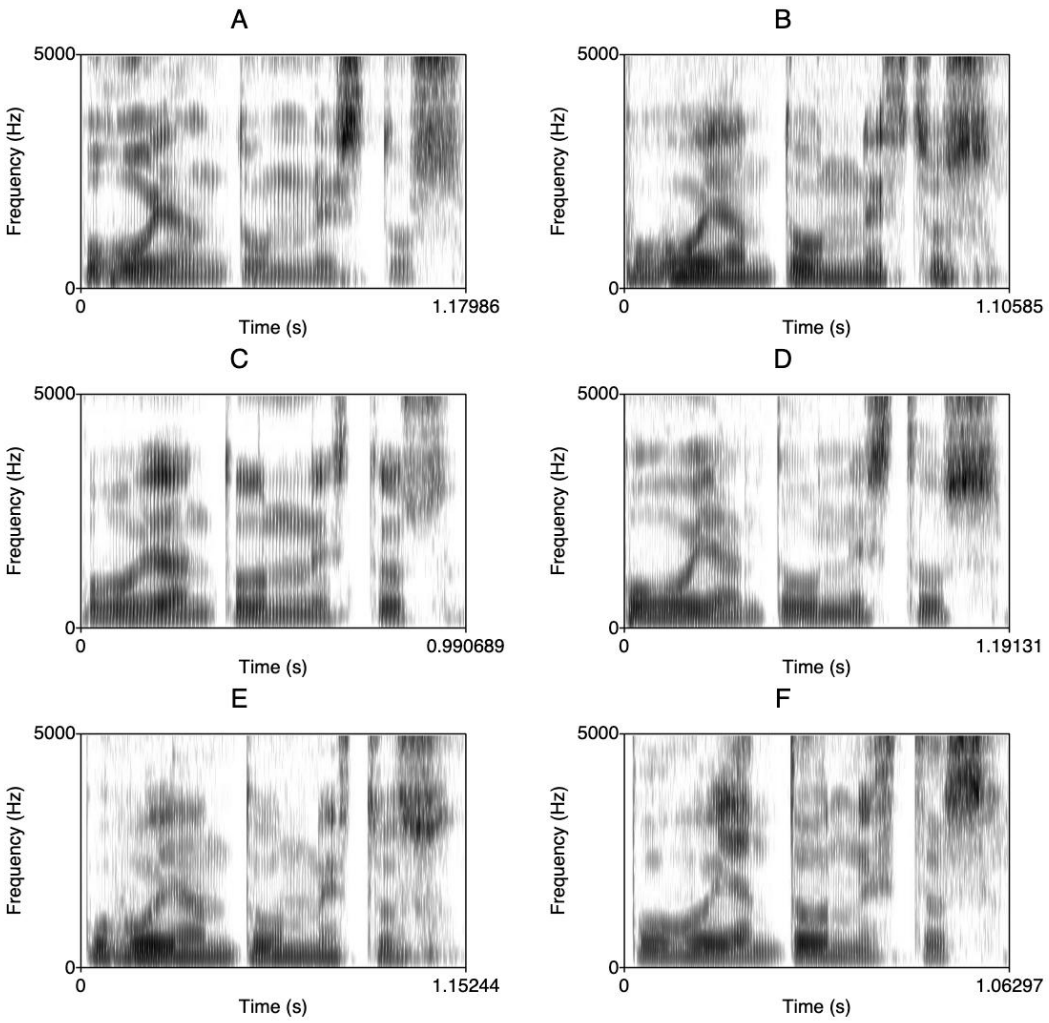
It is possible to represent it as an image where rows=frequencies, columns=different windows d, pixel intensities=$|X_d(f)|$





Final Fantasy VIII – Compression of Time

SPECTROGRAMS-EXAMPLES



5 male speakers uttering "puhujantunnistus" (in finnish: speaker recognition).

Which are the same?

DFT FOR IMAGES

Continuous

$$I_a(f_x, f_y) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} i_a(x,y)\, e^{-j2\pi(\mathrm{f}_x x + \mathrm{f}_y y)} dx dy$$

with its inverse

$$i_a(x,y) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} I_a(f_x, f_y)\, e^{j2\pi(\mathrm{f}_x x + \mathrm{f}_y y)} df_x df_y$$

Discrete

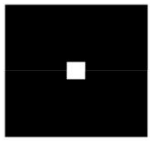
$$I(u,v) = \sum_{x=0}^{N-1} \sum_{y=0}^{M-1} i(x,y) e^{-j\frac{2\pi}{NM}(ux+vy)}$$

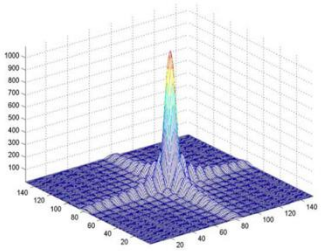
with its inverse

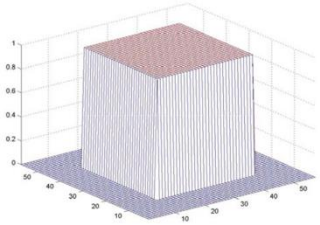
$$i(x,y) = \frac{1}{NM} \sum_{x=0}^{N-1} \sum_{y=0}^{M-1} I(u,v) e^{j\frac{2\pi}{NM}(ux+vy)}$$

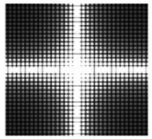
2D DFT: EXAMPLES

Rect 2D

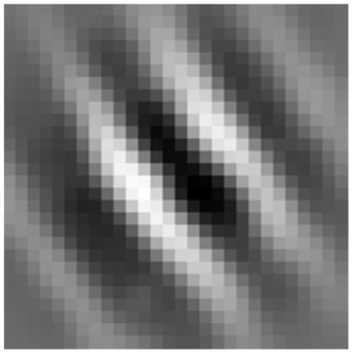


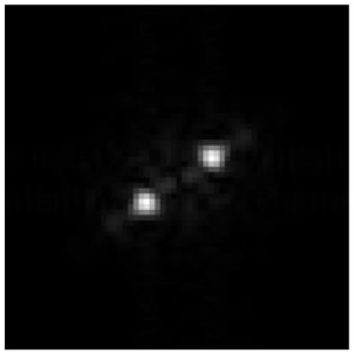






Fingerprint detail

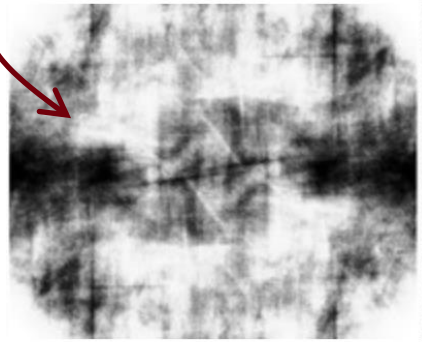




$$w(x,y) = A\cos\left(2\pi f_0(x\cos\theta + y\,sin\theta)\right)$$



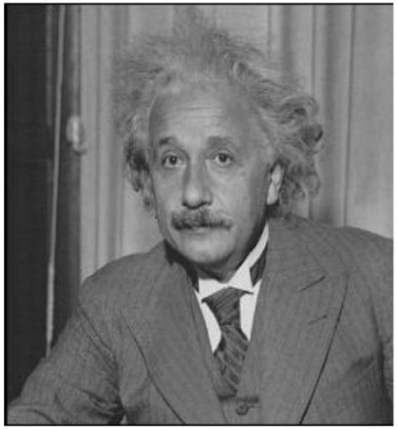
magnitude

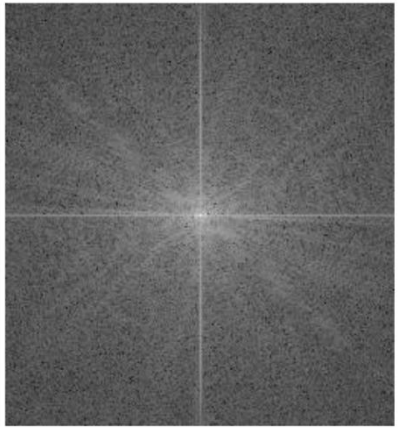


phase



FFT





Log-magnitude of spectrum

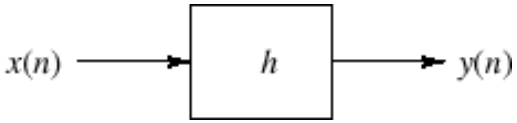
FILTERING

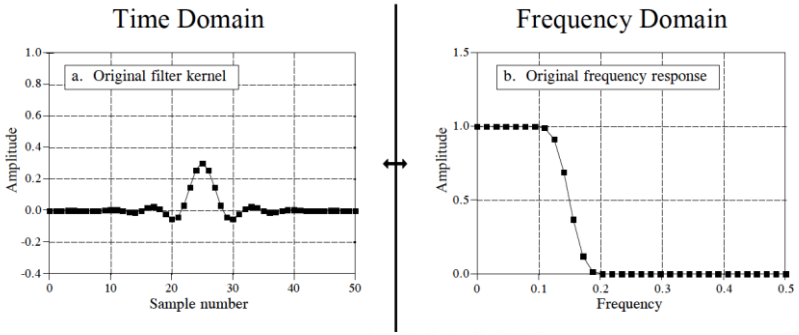
FILTERING IN FREQUENCY DOMAIN

Filtering in time =.samplewise multiplication in frequency

$H(\omega) = DFT(h)$
$X(\omega) = DFT(x)$

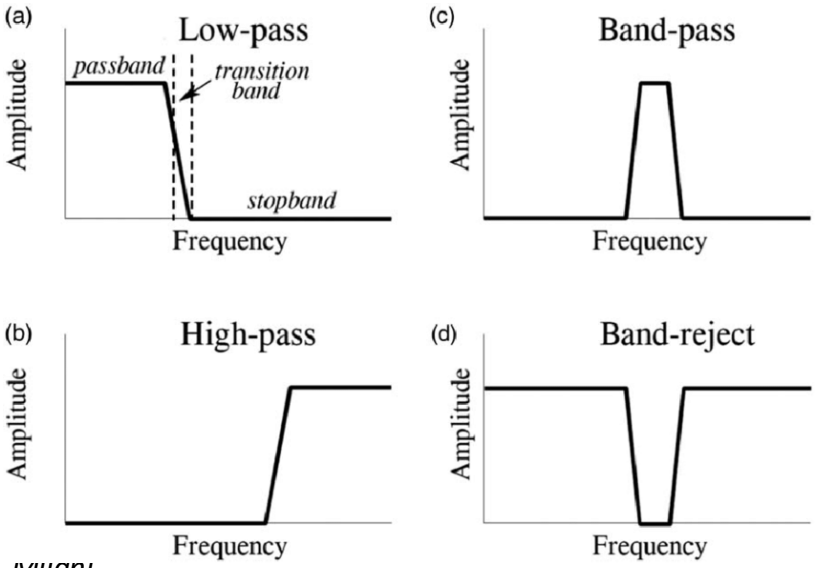
$Y(\omega) = H(\omega)X(\omega)$





Usually amplitude response $H(\omega) \in \mathbb{C}$ and can be represented in abs $|H(\omega)|$ and phase $\angle H(\omega)$

Amplitude usually measured in dB. = $20\ log_{10}\ |H(\omega)|$. *Phase in radiants*



EXAMPLES OF FILTERING

Filter responses are therefore maximum whenever input signal and response are aligned and very similar.

i.e. filters catch some specific structure in a signal

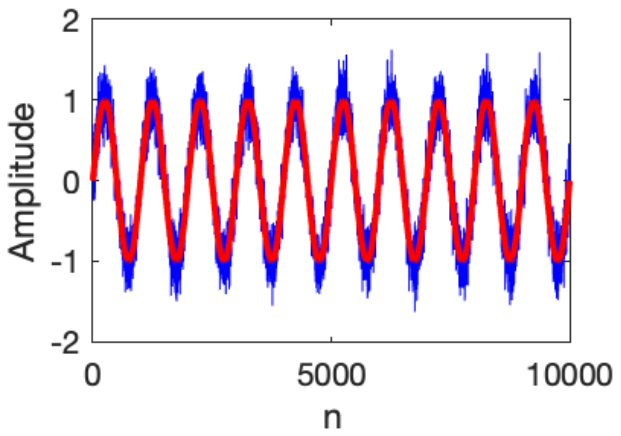
E.g. Average filters

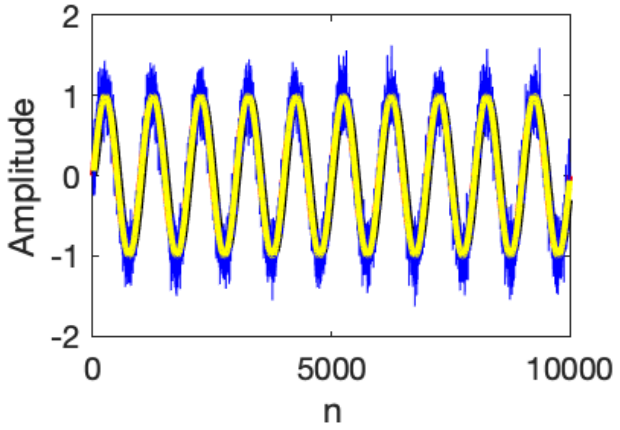
$$x_0[n] = \sin\left(\frac{2\pi n}{100}\right)$$

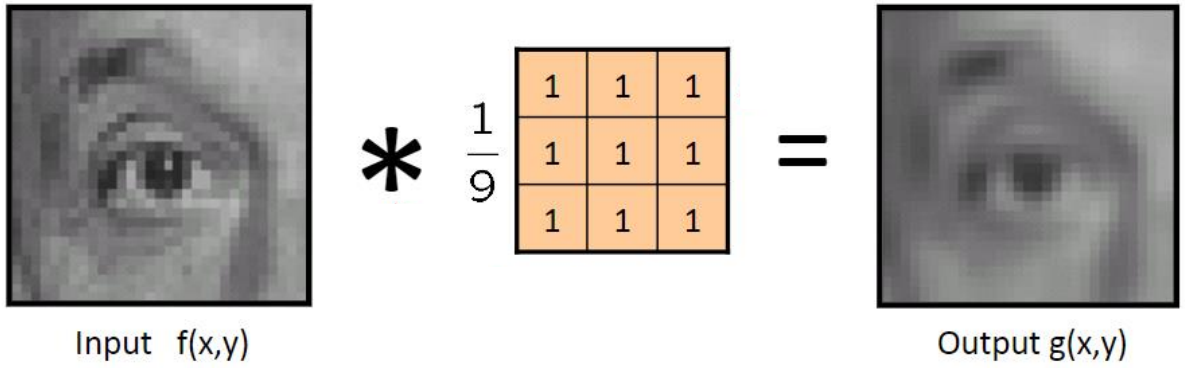
$$x[n] = x_{0[n]} + e[n]$$

AWGN

$$y[n] = h[n] * x[n]$$





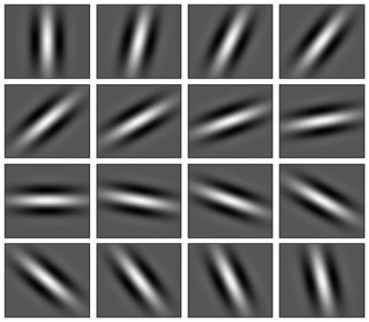


FILTERING FOR BIOMETRICS

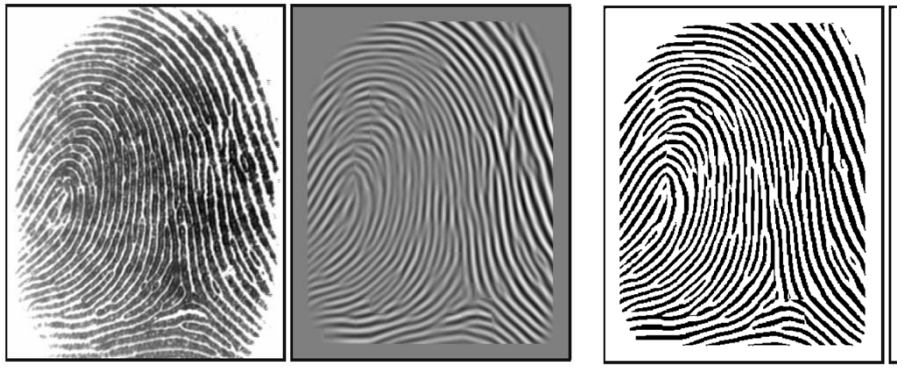
Filtering is very helpful in **ridge extraction** for fingerprints

a) pores are brighter
b) ridges can be broken due to cuts or creases;
c) adjacent ridges may appear joined due to moist skin or pressure.

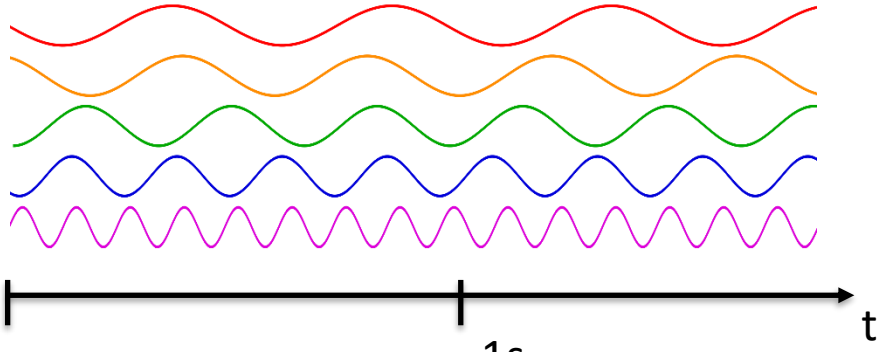
Gabor filters



Before Gabor filter | After Gabor filter | Thresholded

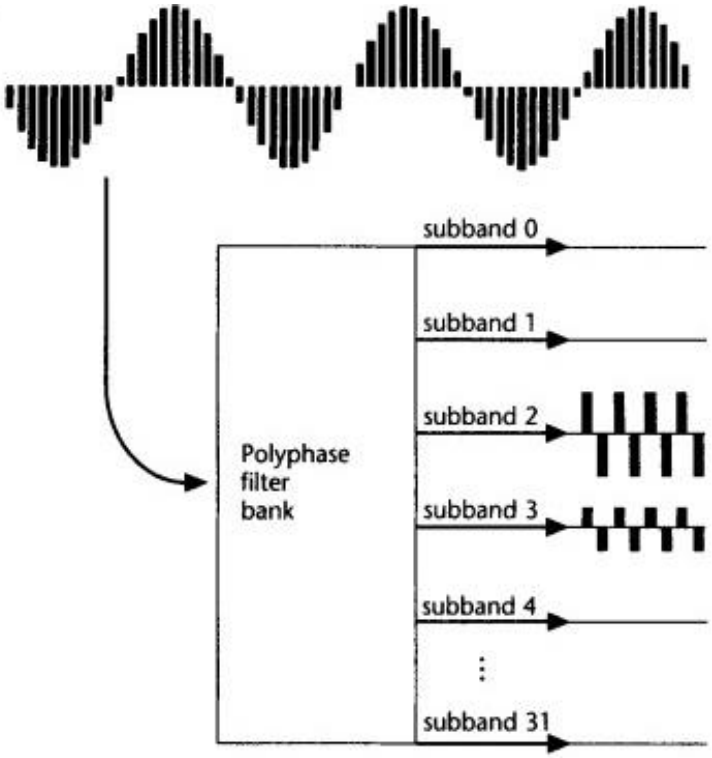


Also used for **audio,** selection of specific frequencies - denoising





SUBBAND DIVISION

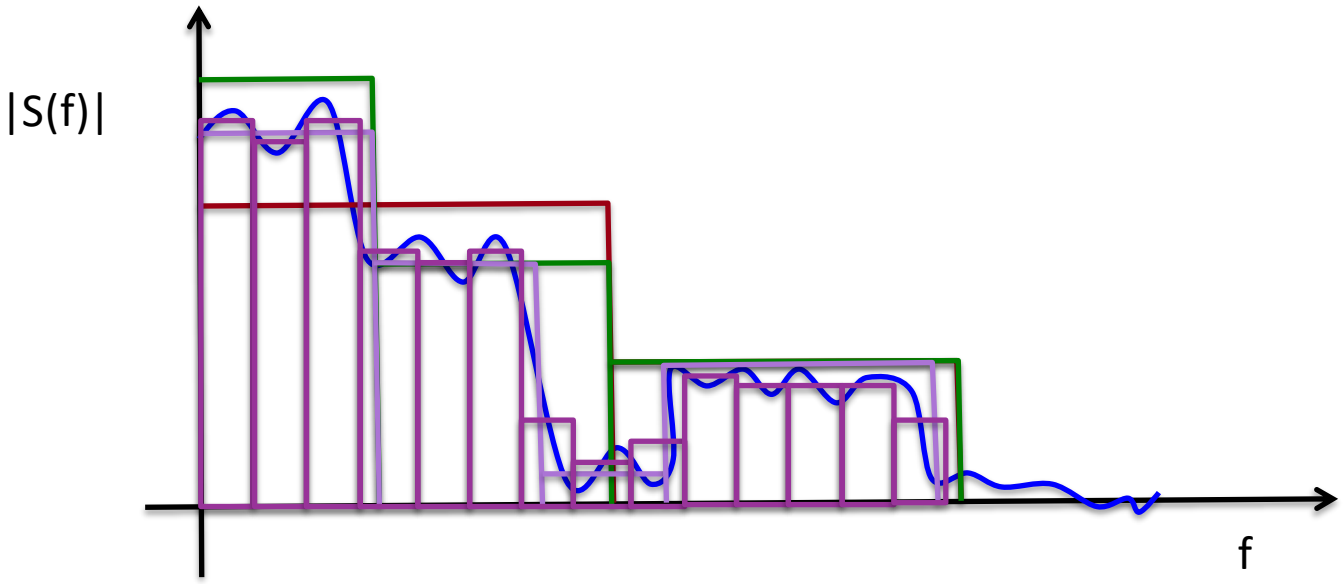


- Subband coding is a way of encoding a signal/image by breaking it down into the sum of several simpler components.
- The original signal is filtered and each frequency subband is isolated.
- Each component is then coded independently.

Images:
Each component can be related to the details of an image

Sound:
Each component can be related to different objects, people, or refinement in signal characterization.

EXAMPLE



If the frequency ranges are small enough, the accuracy of the representation can be high.

How many intervals? Depends on the signal

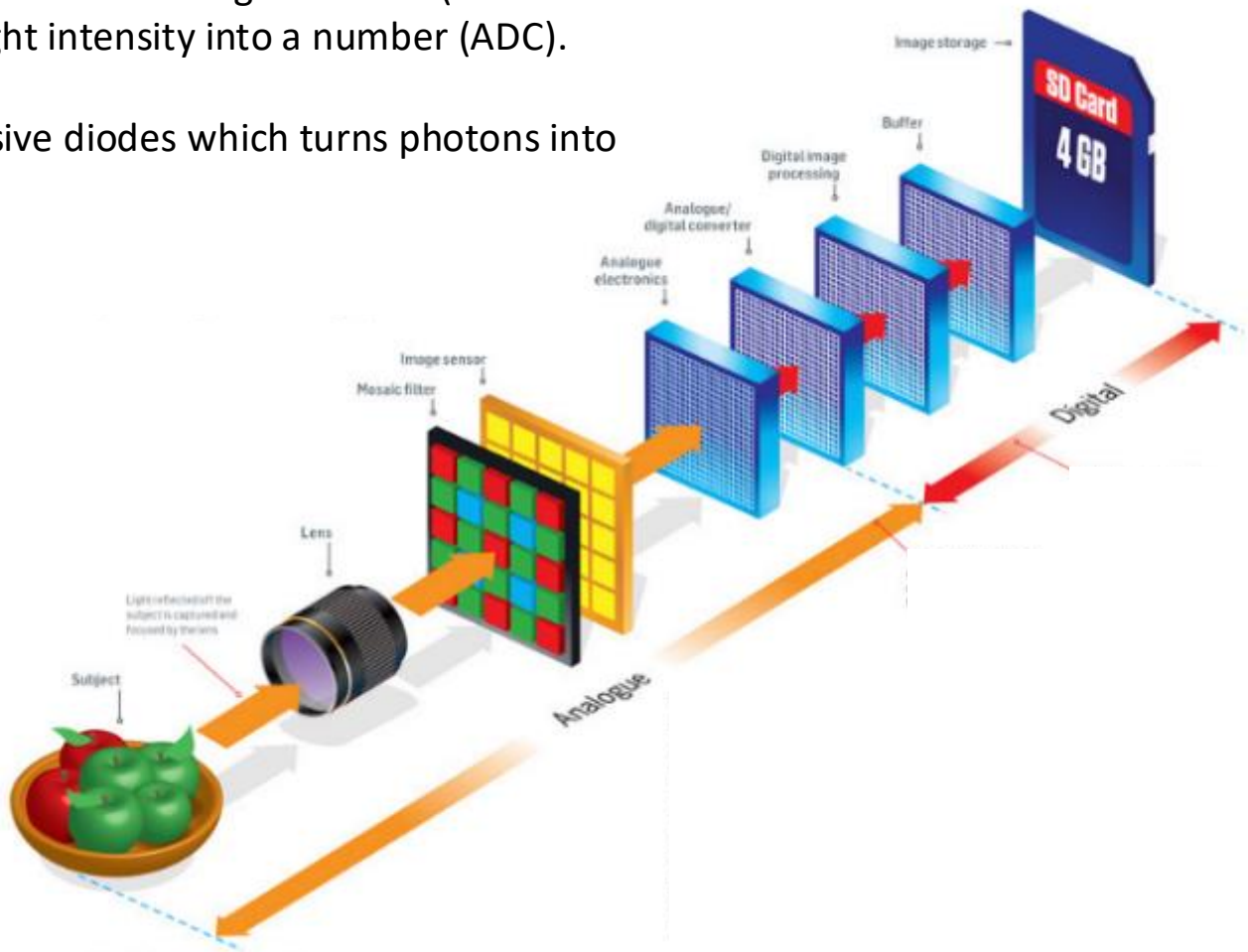
original | 128 bands | 512 bands | 1024 bands | 8192 bands | 64 bands + impulsive noise | 64 bands + Hallelujah

"And the only word there spoken was the whispered word, `Lenore?'" (The Raven, E.A.Poe)

DIGITAL CAMERAS

DIGITAL CAMERA MODEL

- Projective model is applied from object points to analog optics
- Image plane (real) is associated to a matrix of finite elements (pixel): each element is a digital sensor (CCD o CMOS) that converts light intensity into a number (ADC).
- Matrix of photoresponsive diodes which turns photons into electrons

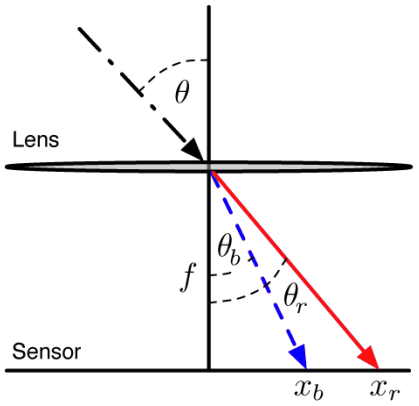


LENS ABERRATION

Most images contain a variety of aberrations that result from imperfections and artifacts of the optical imaging system

Fresnel's equation. $n \sin\theta = n_f \sin\theta_f$

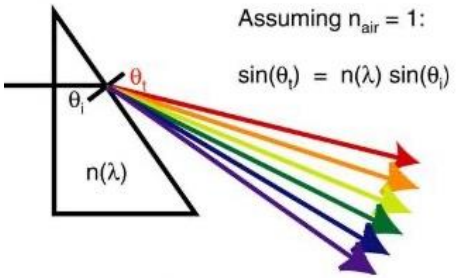
θ : incident angle
θ_f: refraction angle
n, n_f: refraction indexes. (depends on wavelength λ)



Different colors = different λ = different refractions. **Colors separate!**

$n \sin\theta = n_B \sin\theta_B = n_R \sin\theta_R$

Since n are different, angles must compensate $\theta_R \neq \theta_B$



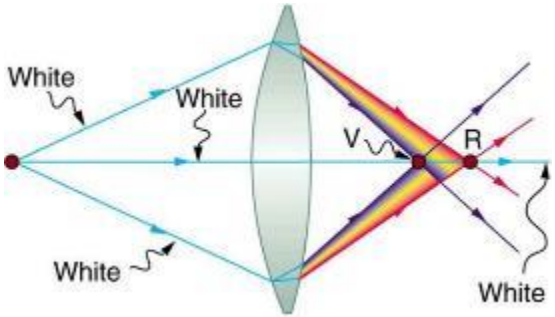
Dividing by $\cos\theta_R \approx \cos\theta_B$ we have

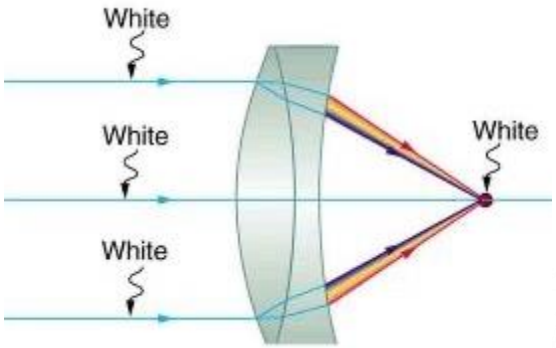
$$n_B \operatorname{tg}\theta_B = n_B \frac{x_B}{f} \approx n_R \frac{x_R}{f} = n_R \operatorname{tg}\theta_R$$

$$x_R \approx \frac{n_B}{n_R} x_B = \alpha x_B$$

CHROMATIC ABERRATION

Different wavelengths converge to different points





Compensated (achromatic doublet)

Near Lens Center



Near Lens Outer Edge



CHROMATIC ABERRATION

- From previous equations we have. $(x_R, y_R) \approx \alpha(x_B, y_B)$

- A more complex model add a center of aberration

$$x_R = \alpha(x_B - x_0) + x_0$$
$$y_R = \alpha(y_B - y_0) + y_0$$

Using green as reference find parameters α, x_0, y_0

RADIAL DISTORTION



original



corrected

All the lenses show this (compensated in the firmware)





Utterly evident for fish-eye lenses



For omnidirectional cameras, use together with tangential distortion

LENS FORENSICS

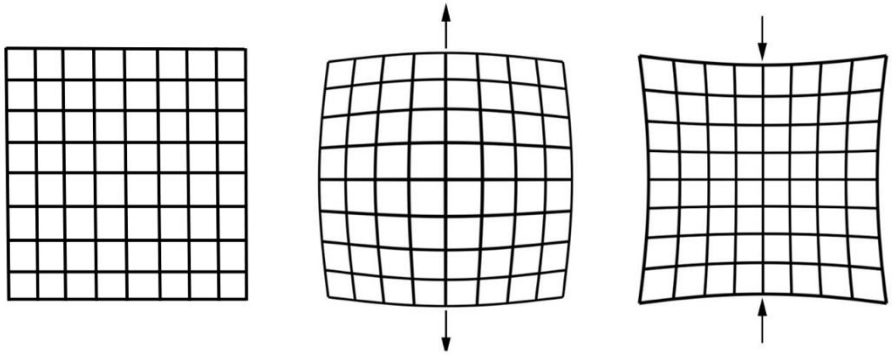
Most lenses introduce different kinds of lens aberrations such as spherical aberration, field curvature, lens radial distortion and chromatic distortion.



a



b

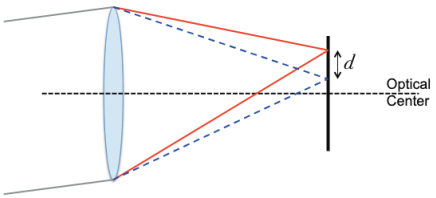


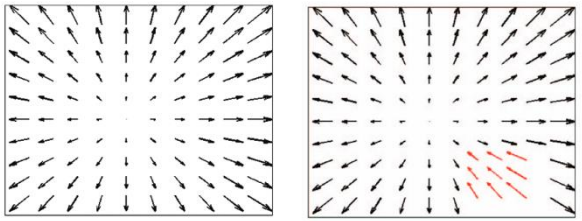
a b c

Aberrations of sphericity are due to the building process: calibrating the camera it is possible to estimate distortion parameters. → features for identification

$$r_u = r_d + k_1 \cdot r_d^3 + k_2 \cdot r_d^5$$

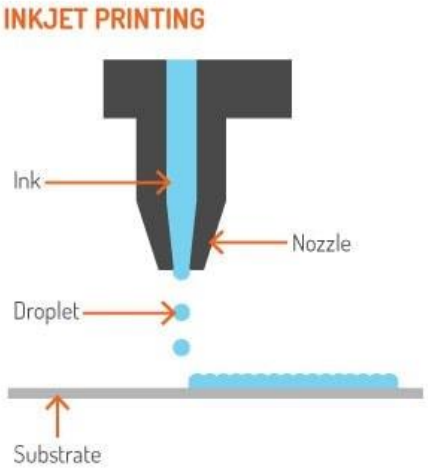
Similar considerations for colors

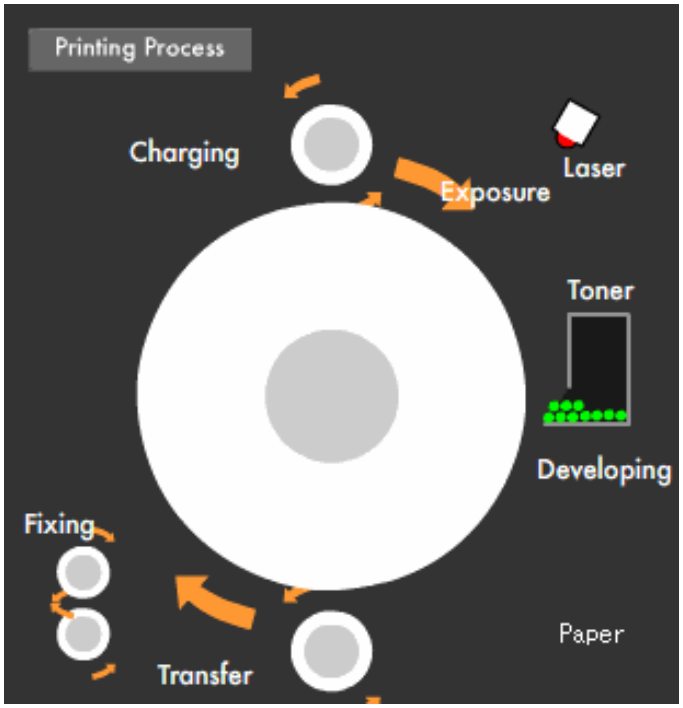




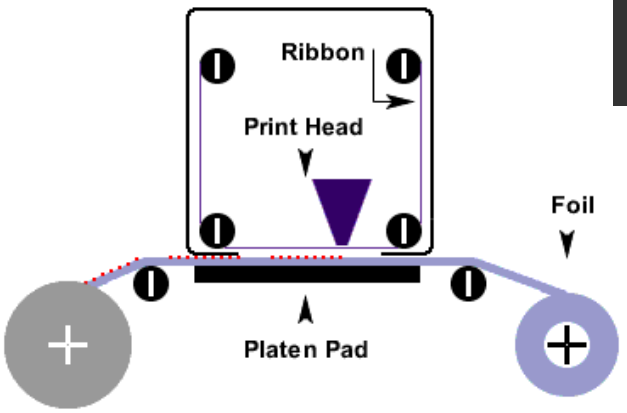
K. S. Choi, E. Y. Lam, and K. Wong. Source camera identification using footprints from lens aberration. In Proc. SPIE, Electronic Imaging, Media Watermarking, Security, and Forensics, volume 6069, pages 60690J–60690J–8, 2006.

PRINTER AND SCANNER FORENSICS



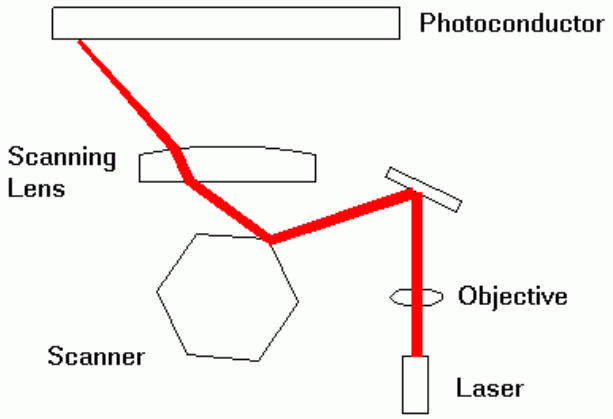


Laser printers



Inkjet printers

Laser scanning



A QUICK OVERVIEW

Laser printers

- use a laser to draw image as an electrostatic pattern on a rotating drum.
- Ink (powder) sticks to charged areas
- transfers from the drum onto paper.
- heat and pressure melt the toner into the paper

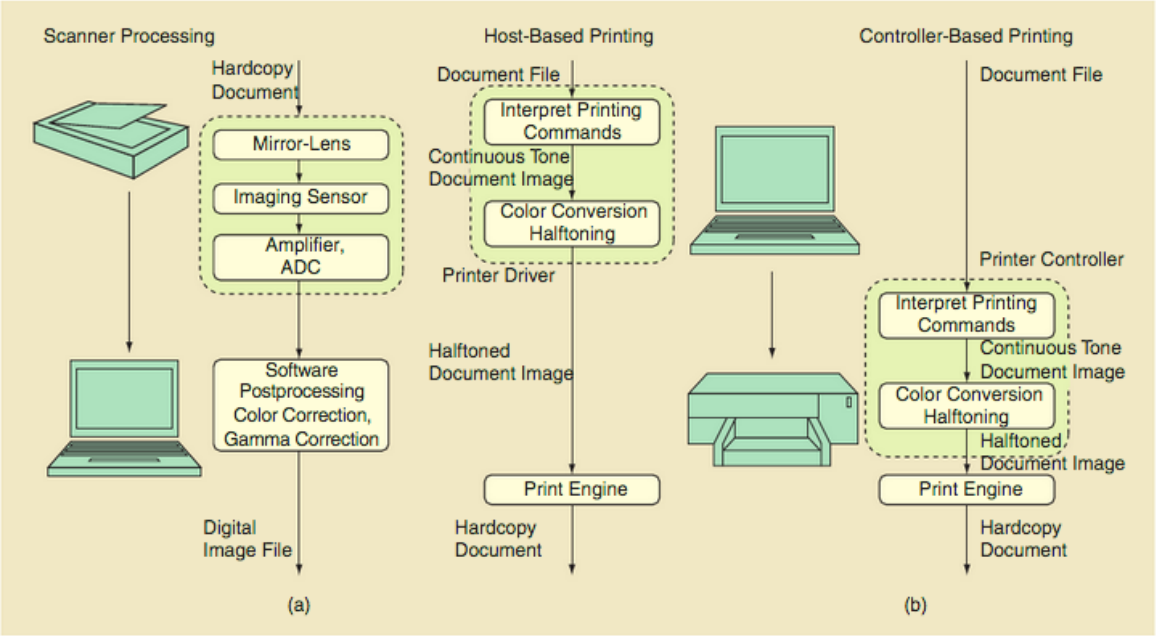
Ink-jet printers

- Fine nozzles spray tiny droplets of ink onto the paper
- print head moves across the page, placing droplets in precise patterns
- ink dries on the paper.

Scanners

- Light shines onto the document
- sensors measure the reflected light as light moves across the page
- sensors turn measurements into pixels and assembled into a digital image.

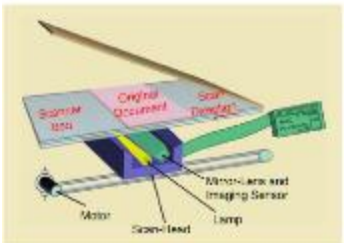
PRINTER AND SCANNER FORENSICS



Chiang et al. "Printer and Scanner Forensics", IEEE Sig.Proc. Mag., 2009

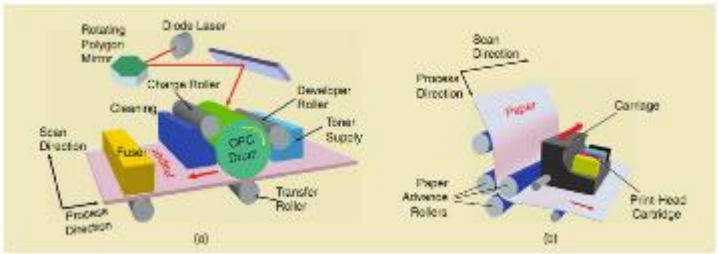
Scanners' signatures:

- Sensor noise
- Motor and timing
- Colors and bands
- Lens, glass, mirrors (scratches, smudges, …)



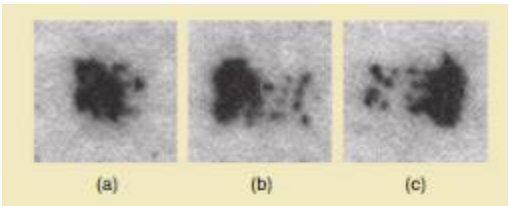
Laser printers' signatures:

- gear mechanism,
- polygon mirror wobble
- optical photoconductor (OPC) angular velocity



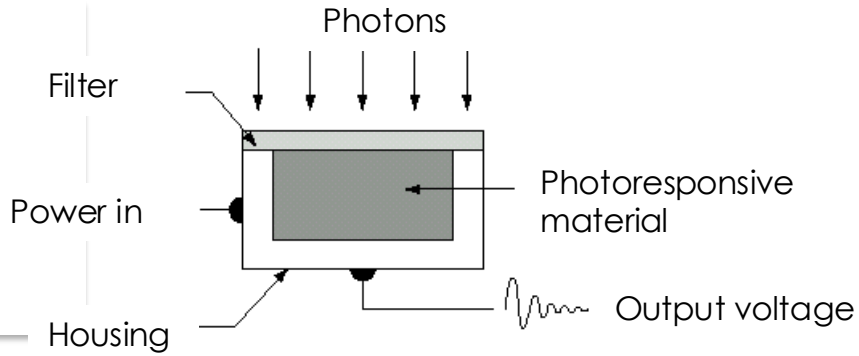
Ink-jet printers' signatures:

- dot shapes
- dot placement
- ink-jet nozzle, ink chemistry,
- periodic variation by missing jets or paper-advance errors,

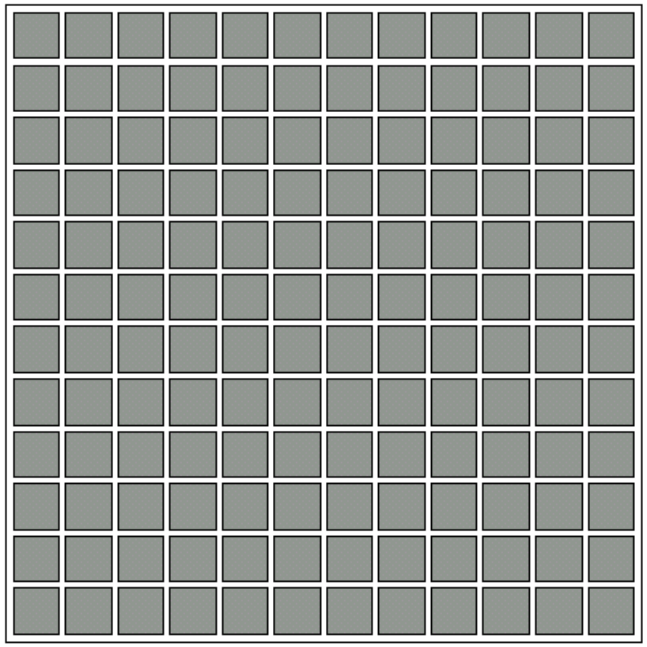


SENSOR NOISE

DIGITALIZATION



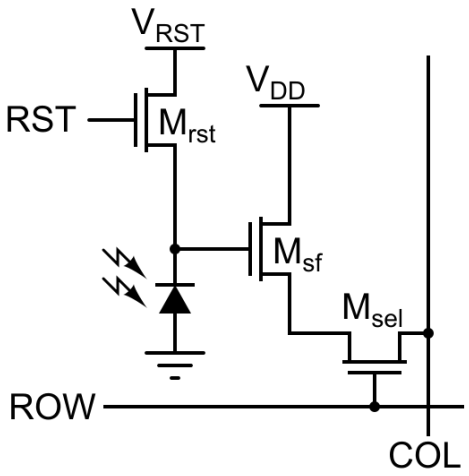




Idea: the energy of incoming photons is converted into a voltage thanks to the photoresponsive material.

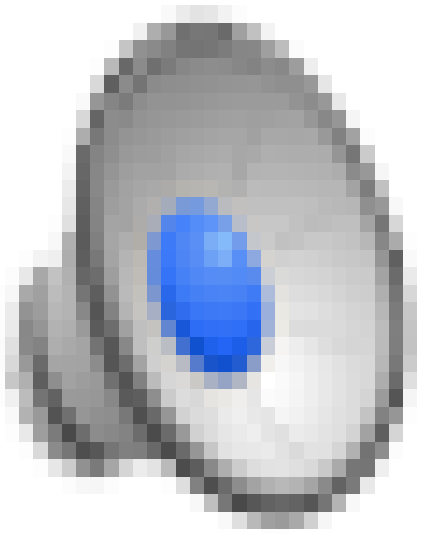
Voltage is proportional to the integral energy absorbed by the sensor (photoelectric effect)

CMOS Active-Pixel Sensors (APS)

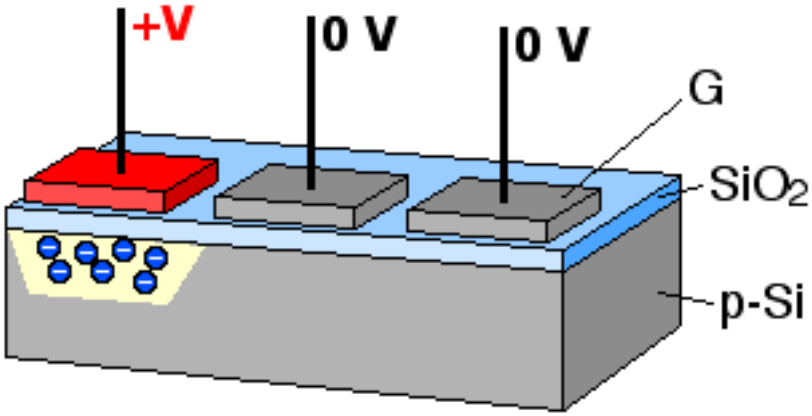


1. Every pixel = photodiode + readout amplifier
2. This allows converting the energy stored as charge in the photodiode in tension and amplify it; the value is then sequentially read according to a row-column ordering.
3. 1 pixel = 3 transistor: conversion electronic charge/ tension, reset of photodiode, reading.
4. Sensor array is organized in a checkerboard structure; dedicated circuitry is used to control the sequential reading.
5. Address can be specified with (x,y) coordinates

CMOS APS



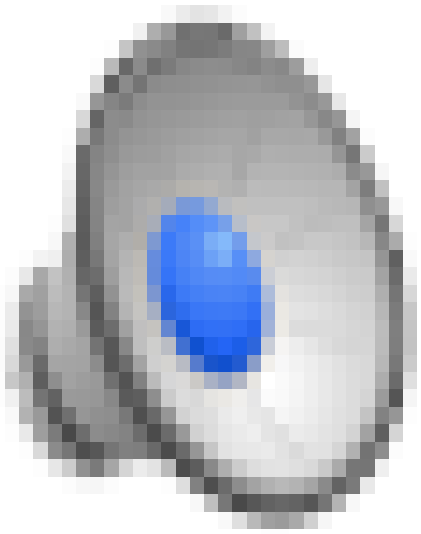
CCD



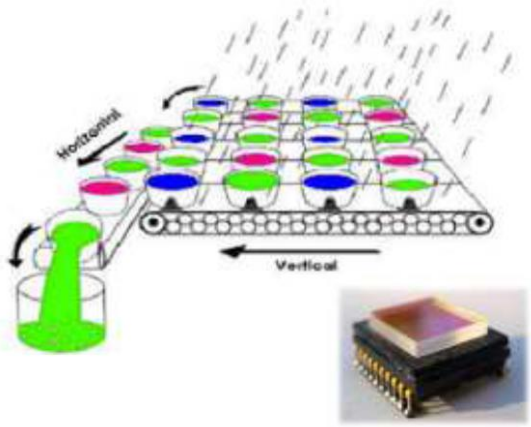
- Two components
- Photoresponsive region
- Shift register to transmit the acquired value(CCD).
- Pixel acquisition:

1. Light hits the capacitor of the photoresponsive region;
2. Capacitor starts charging: the amount of charges depend on the intensity;
3. When acquisition time ends, the charge is transferred to the nearby capacitor (shift register) using a control circuitry.
4. The last capacitor is discharged using an amplifier that change the charge in a voltage. Voltage is sampled and memorized.

CCD



PHOTOSENSORS IN A CAMERA



Similar to buckets collecting rain (photons) until a certain level (the pixel value) is reached

Ideally

When uniform light falls on a camera sensor, each pixel should output exactly the same value.

Practically

Small variations in cell size and substrate material result in slightly different output values

Some noise is introduced into the image

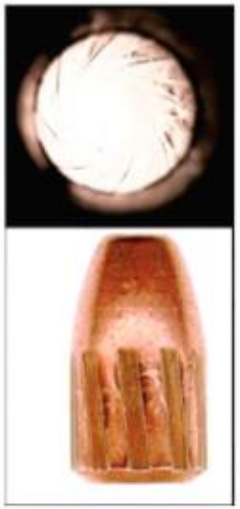
- Fixed Pattern Noise (FPN)
- Photo Response Non Uniformity (PRNU)

CAMERA BALLISTIC







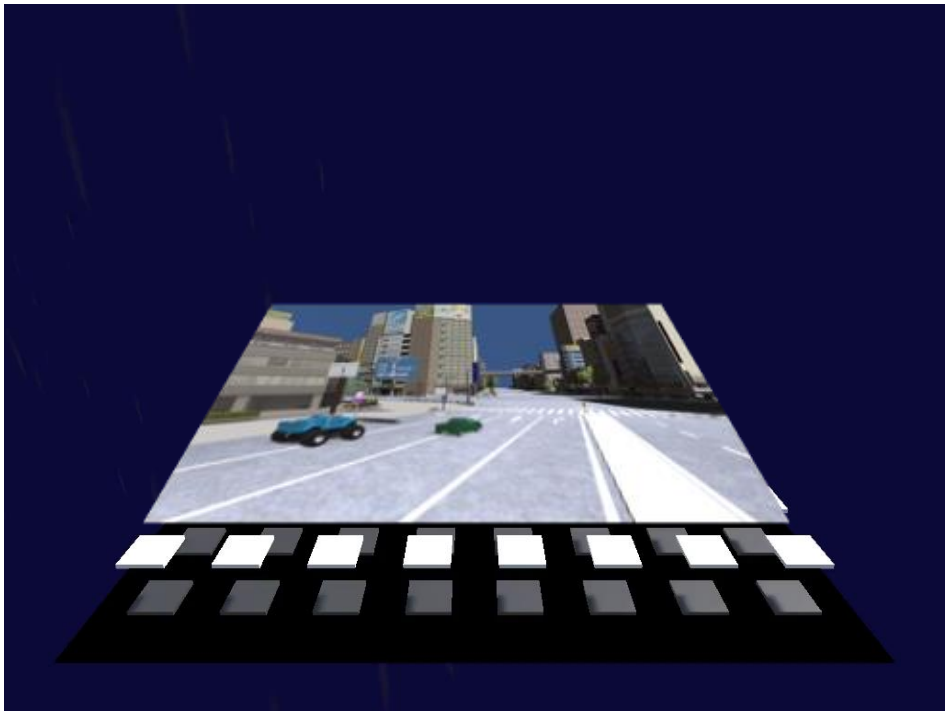


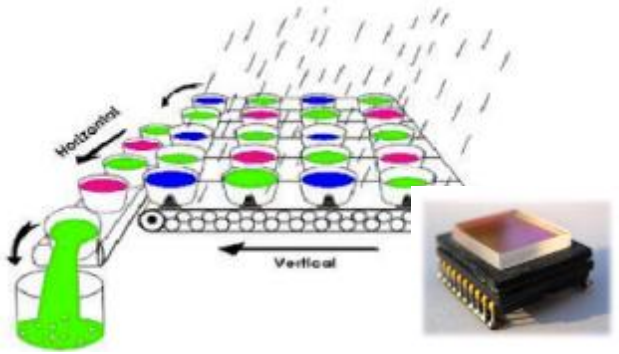
Like a gun on the bullet, the acquisition devices leave distinctive traces on the acquired content

What for?

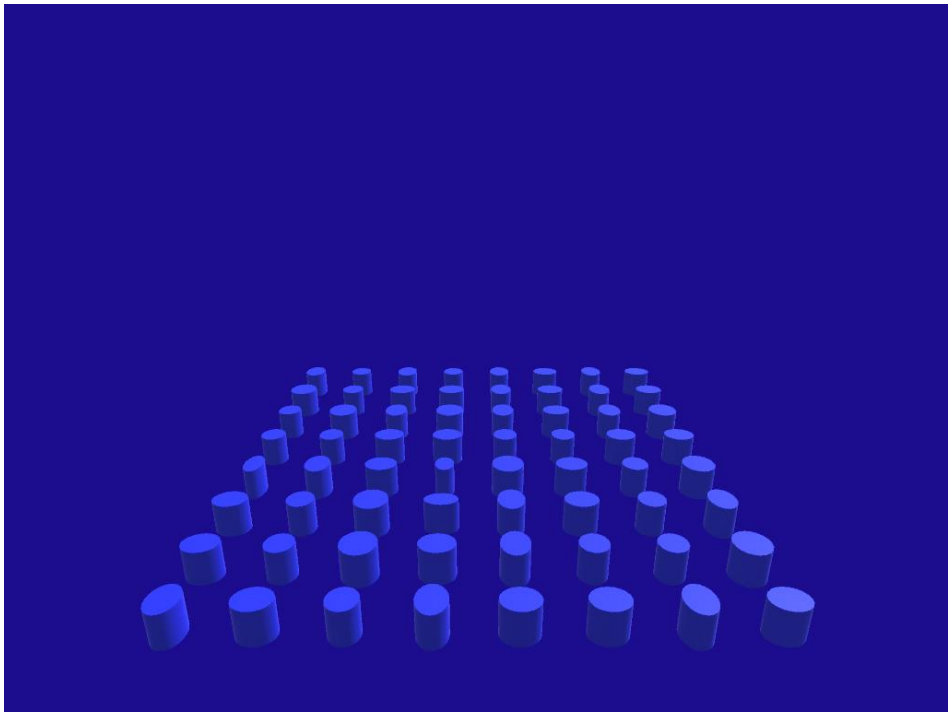
- distinguish if the content is real or computer generated
- distinguish between different devices (camera, scanner, etc.)
- distinguish which camera captured the image: **Camera identification**

PHOTO RESPONSE NON UNIFORMITY (PRNU)









PRNU (PHOTO-RESPONSE NON-UNIFORMITY)

PRNU is caused by the different sensitivity of the sensors to light

Due to the manufacturing process – does not depend on temperature and time

$$I = I_0 + I_0 K + \Theta$$

$\boldsymbol{I_0}$ sensor output without noise

K PRNU fingerprint of camera C **(multiplicative noise)**

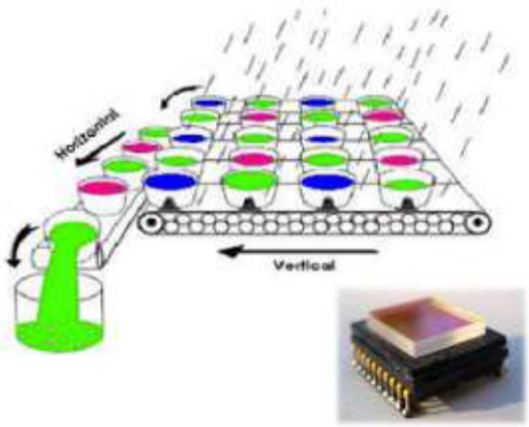
$\boldsymbol{\Theta}$ all the others noise terms (shot, readout, etc.)

A more extended modelling

$$I = g^{\gamma}[I_0(1 + K) + \Theta]^{\gamma} + Q$$

where Q gathers compression/quantization noise and g is the gain factor

PHOTOSENSORS IN A CAMERA



Similar to buckets collecting rain (photons) until a certain level (the pixel value) is reached

Ideally

When uniform light falls on a camera sensor, each pixel should output exactly the same value.

Practically

Small variations in cell size and substrate material result in slightly different output values

Some noise is introduced into the image

- Fixed Pattern Noise (FPN)
- Photo Response Non Uniformity (PRNU)

PRNU (PHOTO-RESPONSE NON-UNIFORMITY)

Expand — Add and subtract

$$W_I = I - \hat{I}^{(0)} = I - \hat{I}^{(0)} + I \cdot K - I \cdot K =$$

$$= I \cdot K + I^{(0)} - \hat{I}^{(0)} + (I^{(0)} - I)K + \Theta$$

Σ

Independent from IK

PRNU estimation

① Denoise the image $\hat{I}^{(0)} = F(I^{(0)})$

② Compute difference $W_I = I - \hat{I}^{(0)} = IK + \Sigma$

Repeat 1 and 2 for many images $I_1, I_2, \ldots, I_N$ generating $W_1, W_2, \ldots, W_N$

③ $\hat{K} = \dfrac{\sum_{i=1}^{N} W_i I_i}{\sum_{i=1}^{N} (I_i)^2}$ Compute noise

PRNU DETECTION

Recipe for a good estimate of PRNU

- The estimation is done on flat-field images (uniform content, low variance)
- The luminance of the images should be as high as possible but not saturated
- About 20 flat-field (bright) images suffice to have a good estimate of K

Verification

- Let J be another image from the same camera (or another one)
- The presence of K can be determined by a correlation detector, i.e.,

$$\boldsymbol{corr}(\boldsymbol{W_J}, \widehat{\boldsymbol{K}} \cdot \mathrm{J})$$

NCC (Normalized Cross-Correlation)

$$\boldsymbol{\rho} = \frac{(\boldsymbol{X} - \bar{\boldsymbol{X}}) \cdot (\boldsymbol{Y} - \bar{\boldsymbol{Y}})}{\|\boldsymbol{X} - \bar{\boldsymbol{X}}\| \cdot \|\boldsymbol{Y} - \bar{\boldsymbol{Y}}\|}$$

$$X \cdot \mathrm{Y} = \sum_{i,j} X[i,j] \cdot Y[i,j]$$

$$\|X\| = \sqrt{X \cdot \mathrm{X}}$$

PCE (Peak-to-Correlation Energy)

$$\mathbf{PCE} = \frac{\boldsymbol{\rho}^2}{\frac{1}{\boldsymbol{MN}} \Sigma_{\boldsymbol{i}} \Sigma_{\boldsymbol{j}} \boldsymbol{NCC_{i,j}}^2}$$

$$NCC_{i,j} = \frac{\sum_{k=0}^{m-1} \sum_{l=0}^{n-1} (X(i,j) - \bar{X}) \cdot (Y(k+i, l+j) - \bar{Y})}{\|X - \bar{X}\| \cdot \|Y - \bar{Y}\|}$$

PRNU IN PRACTICE

